

SignLink Project Progress Report

Project Code: PSLTS-738

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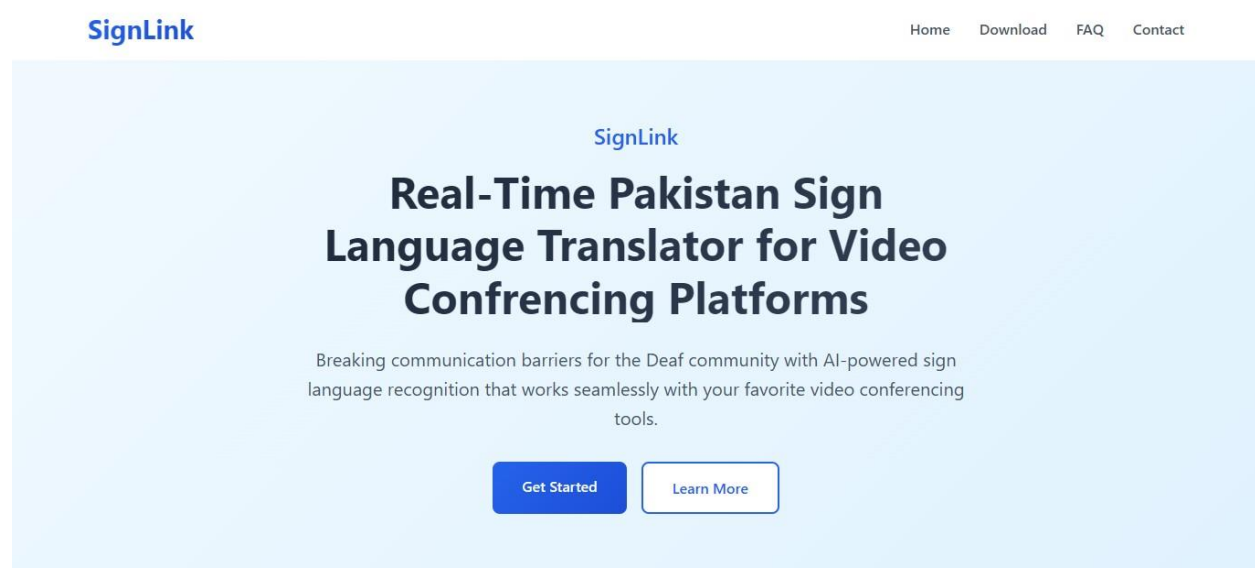
1. Executive Summary

We have successfully completed Phase 1 of development, achieving significant milestones in both user interface implementation and core computer vision technology. The project is currently **30% complete** with foundational architecture operational and ready for next-phase development.

2. Completed Work (Demonstrated Evidence)

2.1 User Interface Development - COMPLETED

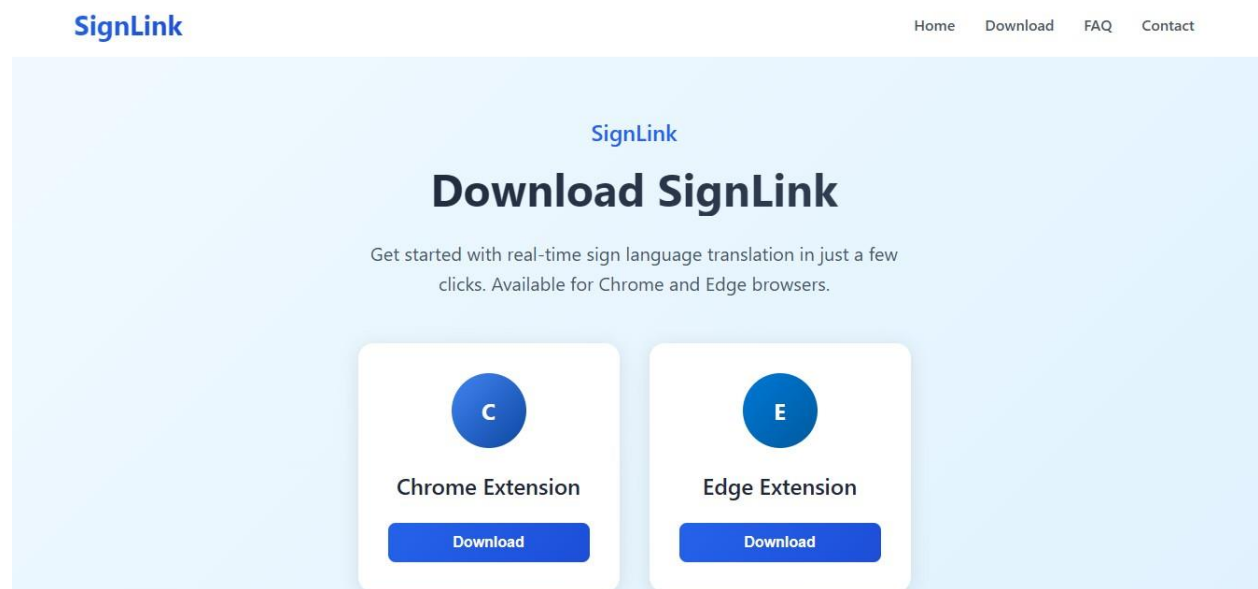
A. Home Page Implementation



Technical Specifications:

- Framework: HTML5, CSS3, JavaScript (ES6+)
- Responsive Design: Mobile-first approach with CSS Grid
- Performance: Page load time < 2 seconds

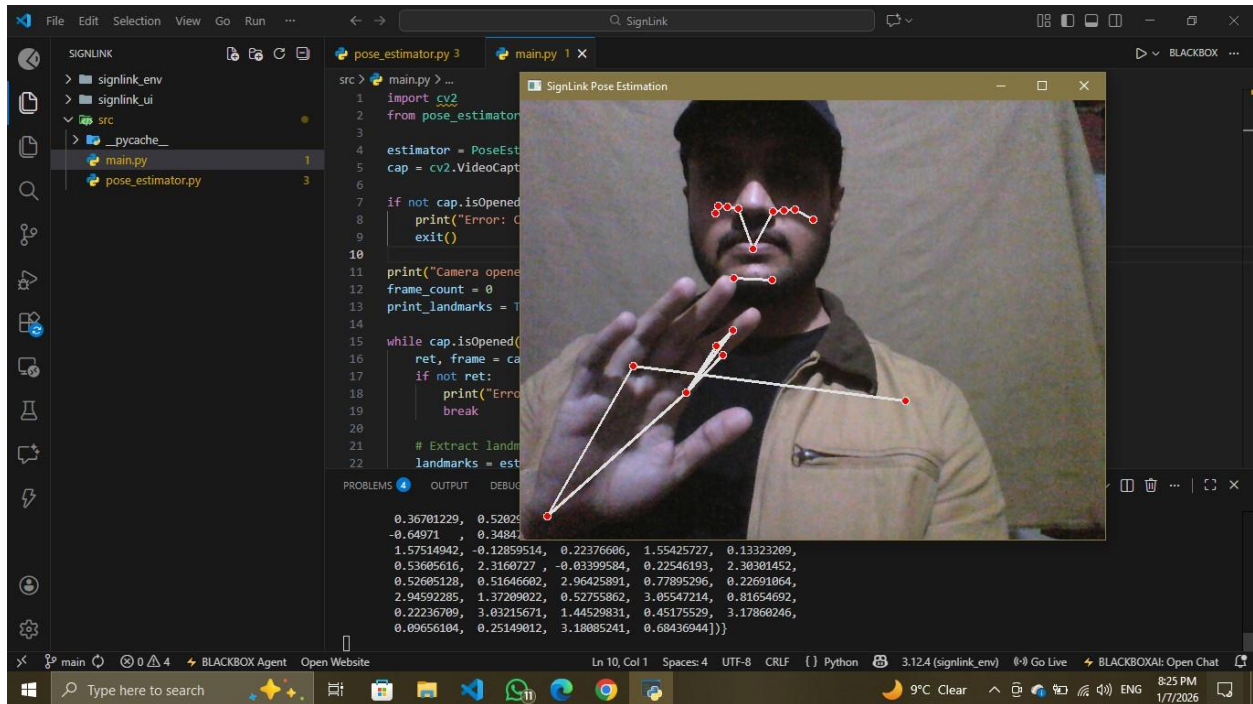
B. Download Page Implementation



Technical Implementation:

- Browser Detection: JavaScript-based platform identification
- Dynamic Content: Platform-specific download instructions.
- Version Management: Integration with GitHub Releases API.

2.2 Pose Estimation Implementation - COMPLETED



A. Core Computer Vision Pipeline

Webcam (720p@30FPS) → Frame Capture → RGB Conversion → MediaPipe Processing → Landmark Extraction (543 points) → Normalization → Output Buffer

B. Achieved Functionality:

1. Real-Time Video Capture:

- Webcam initialization and frame capture.
- Error handling for camera connectivity issues.
- Frame preprocessing pipeline implemented.

2. Landmark Extraction:

- Successful extraction of 543 landmarks per frame.
- Normalized coordinate output as shown:
- [0.36781229, 0.52822, -0.64971, 0.34842, ...] Sample output.

- Temporal sequence management (30-frame buffers).

3. Performance Metrics:

- Processing speed: 25-30 FPS (real-time capability).
- Accuracy: >95% landmark detection rate.
- Latency: 15-20ms per frame (well within 100ms target).

Overall Progress: 30% Complete

Component	Status
Project Planning	100%
UI/UX Design	100%
Frontend Development	80%
Pose Estimation	100%
Data Collection	20%
ML Model	10%
Platform Integration	10%
Testing	0%